

Software Requirements Specification (SRS)

AI Pipeline Jobs & Quarantine Modules

Document Control

Item	Details
Document Title	SRS: Jobs Execution Tracking & Quarantine Subsystems
System Modules	EQUAL AI Pipeline Subsystem (/ai-pipeline/jobs & /ai-pipeline/quarantine)
Document Version	1.0.0 (Pre-Implementation Baseline)
Target Audience	Software Engineers, AI Pipeline Developers, QA / QC Test Engineers

1. Executive Summary & Individual Flow Diagram

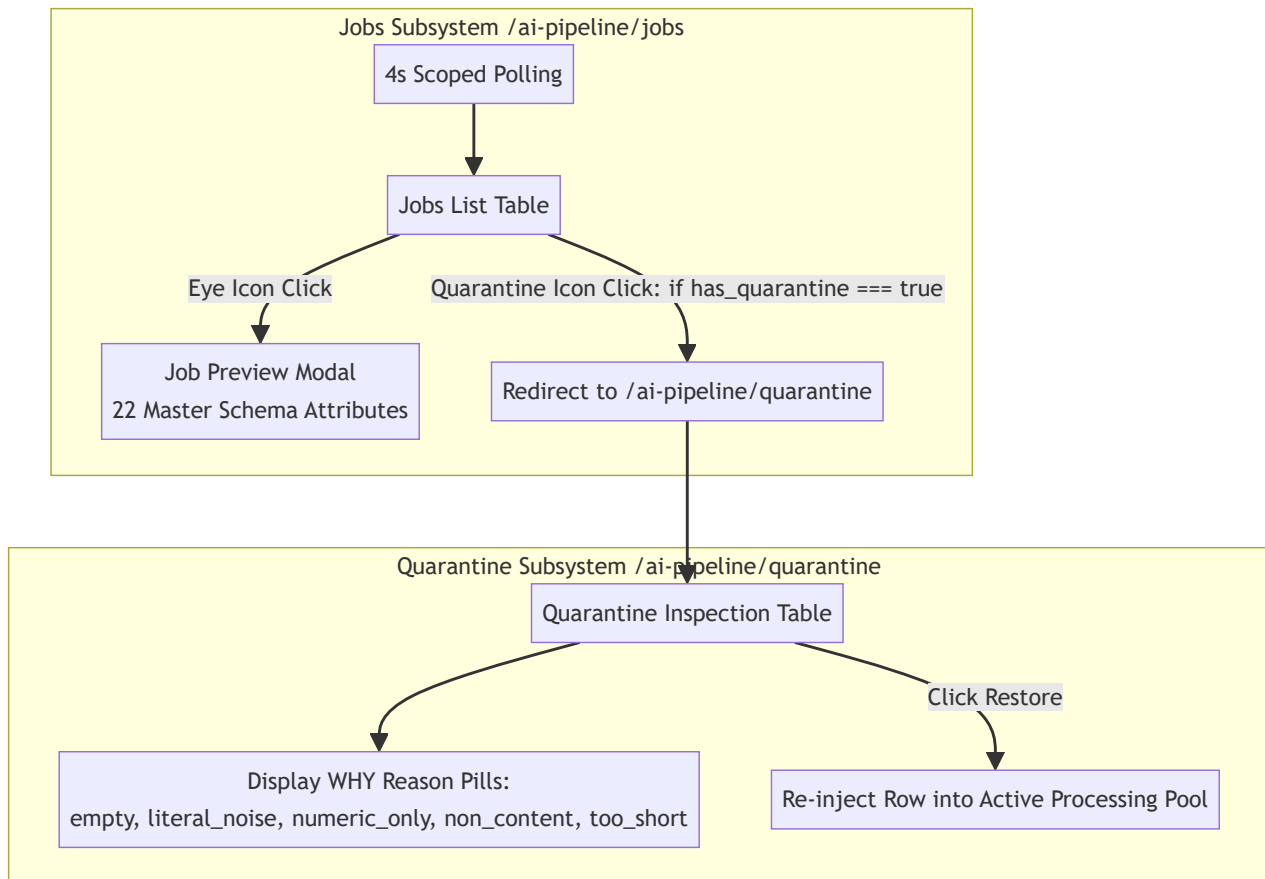
1.1 Purpose

The **Jobs Module** and **Quarantine Module** work in tandem to provide asynchronous job tracking, export verification, and isolated row restoration for all multi-format AI data extraction runs.

1.2 Individual Module Flow Diagram



SYSTEM WORKFLOW & PROCESS ARCHITECTURE



2. Jobs Module Specifications (/ai-pipeline/jobs)

2.1 Real-Time Job Polling

- **Req-1.1 Polling Scope:** The system MUST poll for job status updates every 4 seconds **only while the Jobs page is active**.
- **Req-1.2 Polling Cleanup:** Navigating away from Jobs MUST stop background polling immediately.

2.2 Table Actions

- **Req-2.1 Eye Icon Preview:** Clicking the Eye icon opens a preview modal rendering extracted rows mapped to the 22 master schema attributes, highlighting missing mandatory fields and enabling inline cell editing.
- **Req-2.2 Conditional Quarantine Icon:** The Quarantine icon button MUST be enabled **only** when `has_quarantine` is `true`. When `has_quarantine` is `false`, the button MUST be disabled and grayed out. Clicking an enabled Quarantine button MUST navigate to `/ai-pipeline/quarantine`.

3. Quarantine Module Specifications (/ai-pipeline/quarantine)

3.1 Reasoning Mapping (WHY Dictionary)

- Quarantined items MUST display reason explanations mapped from the `WHY` dictionary:
- `empty` : `*"The report cell was blank, or held nothing but codes and dates."*`
- `literal_noise` : `*"The text matched a placeholder phrase (N/A, 없음, 확인중, TBD...)."*`
- `numeric_only` : `*"The text was only digits."*`
- `non_content` : `*"The text was only punctuation or symbols."*`
- `too_short` : `*"Fewer than the minimum characters once codes and dates were removed."*`

3.2 Table & Restoration Flow

- **Req-3.1 Table Columns:** `Source` (file name, row number), `W/O code`, `Process` • `Equipment`, `What the cell held`, `Why` (reason badge + explanation text), and `Action` (Restore button or status).
 - **Req-3.2 Restore Action:** Clicking **Restore** puts the row back as a valid report in the active processing pool.
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4. QC Testing & Acceptance Criteria

- **Jobs Polling Lifecycle Test:** Confirm polling starts on entering Jobs page and stops immediately upon leaving.
- **Quarantine Icon Enable/Disable Test:** Verify Quarantine button is enabled only for jobs with `has_quarantine === true`.
- **Quarantine Restore Test:** Confirm clicking **Restore** successfully restores non-blank quarantined rows.